The Islamia University of Bahawalpur

Bachelor of Science in Information Technology



Project Proposal
(SCOPE DOCUMENT)

For

Online Car Rental System
Version 1.0

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Bachelor of Science in Information Technology

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Project Category: (Select all the major domains of proposed project)

○ **A-** Desktop Application/Information System	○ **B-**Web Application/Web Application based Information System	○ **C-** Problem Solving and Artificial Intelligence	○ **D-**Simulation and Modeling	○ **E-** Smartphone Application	○ **F-** Smartphone Game	○ **G-** Networks
○ **H-** Image Processing	○ Other (specify category) _____					

Abstract

The Online Car Rental System is a web-based application that allows users to book cars online for a specific period. The system provides a user-friendly interface that enables customers to view car options, make reservations, and manage their bookings. The application provides real-time availability of cars, and customers can make payments securely online. The system also includes an administrative interface that allows car rental companies to manage their car fleet, track reservations, and generate reports. The Online Car Rental System offers a convenient and efficient way for individuals and businesses to rent cars, eliminating the need for physical visits to car rental agencies. This system provides a reliable and cost-effective solution for customers who require temporary transportation and car rental companies who seek to optimize their operations.

Introduction

The Online Car Rental System is a web-based platform that provides a convenient and efficient way for individuals and businesses to rent cars online. This system enables customers to browse available cars, make reservations, and manage their bookings through a user-friendly interface. The online car rental system offers real-time availability of cars, and customers can securely make payments online.

The online car rental system eliminates the need for customers to visit physical car rental agencies, saving time and reducing the hassle associated with renting a car. Customers can book a car from anywhere with an internet connection and at any time, making it a flexible and convenient option for busy individuals.

The online car rental system also benefits car rental companies by providing them with a centralized platform to manage their car inventory, track reservations, and generate reports. The system optimizes the use of car fleets, and rental companies can efficiently handle multiple bookings simultaneously.

Overall, the online car rental system is a cost-effective and time-saving solution for customers who require temporary transportation and car rental companies who seek to optimize their operations. The system offers a range of benefits, including convenience, flexibility, and real-time availability, making it a popular option for individuals and businesses alike.

Problem Statement

A car rental is a vehicle that can be used temporarily for a fee during a specified period [4]. Getting a rental car helps people get around despite the fact they do not have access to their own personal vehicle or don't own a vehicle at all [1]. The individual who needs a car must contact a rental car company and contract out for a vehicle [1]. This system increases customer retention and simplify vehicle and staff management [4].

1. One of the main problems faced by customers using an online car rental system is the lack of transparency and reliability regarding the availability and condition of the cars. Customers may receiving a car that does not meet their expectations.
2. Another problem faced by customers is the complex pricing structure and hidden fees. Customers may be unaware of additional charges, which can significantly increase the total rental cost.
3. Lastly, customer can face security concerns regarding customer data and payment information. Customers may be hesitant to provide sensitive information online.

Problem Solution for Proposed System

1. Online car rental systems can implement real-time inventory management systems that provide customers with accurate information on car availability and condition.

2. Online car rental systems can implement transparent pricing structures that clearly display all costs associated with the rental, including taxes. Rental companies can also offer flexible pricing options, such as hourly, daily, or weekly rates, to cater to the diverse needs of customers.

3. Online car rental systems protect customer data and payment information. Rental system can also offer secure payment options, such as jazz cash, easy paisa or (hand payment) gateways, to increase customer trust and confidence

Related System Analysis/Literature Review

Different online car rental are currently working like (**Careem)** in the big cities of Pakistan like **Islamabad, Lahore and Karachi.**

As they are working on a small scale and had a minimal protocol of services. And moreover the lack of competitor make the entire organization work under the boundaries and limitations without any facilitation to the customers.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Careem	Poor car management Poor data record about the drivers working with them Lack of on time database updating Need of security improvement	Timely management of the car conditions report The company should kept the proper record of the drivers On time updating in the database make the retired or lefty employs away from the site Routine checking up of the workers may reduce the chances of any incident

Advantages/Benefits of Proposed System

The Online Car Rental System offers numerous benefits for both customers and car rental companies, some of which include:

Convenience: Customers can easily browse available cars and make reservations from anywhere and at any time, eliminating the need for physical visits to car rental agencies.

Cost-effective: Online car rental systems offer competitive pricing and flexible rental rates, enabling customers to save money and choose the most cost-effective option that suits their needs.

Real-time availability: The system provides real-time information on car availability, enabling customers to make informed decisions and choose from a wider range of options.

Efficient: The system streamlines the car rental process, enabling car rental companies to handle multiple bookings simultaneously and optimize their operations.

Transparency: The system provides transparent pricing and information on car conditions, ensuring that customers are aware of all costs associated with the rental.

Customization: The system offers a range of car options, including economy, luxury, and SUVs, enabling customers to choose a car that suits their needs and preferences.

Secure: Online car rental systems offer secure payment options ensuring the privacy and security of customer data and payment information.

Overall, the Online Car Rental System offers a convenient, cost-effective, and efficient solution for individuals and businesses who require temporary transportation.

Scope

This project traverses a lot of areas ranging from business concept to computing field, and required to perform several researches to be able to achieve the project objectives [6].

The area covers include:

- Car rental industry: This includes study on how the car rental business is being done, process involved and opportunity that exist for improvement.
- General customers as well as the company's staff will be able to use the system effectively [2].
- Web-platform means that the system will be available for access 24/7 except when there is a temporary server issue which is expected to be minimal [6].

Modules

Online Car Rental System consists of several modules that work together to provide a seamless and efficient car rental experience for both customers and car rental companies. Here are some of the essential modules of the system:

1. ***User management:*** This module enables customers to register, log in, and manage their accounts. It also allows car rental companies to manage user information, including contact details and booking history.

2. ***Car management:*** This module enables car rental companies to manage their car inventory, including adding, updating, and removing cars from the system. It also allows car rental companies to track car availability and schedule maintenance tasks.

3. ***Reservation management:*** This module enables customers to make online reservations, select pickup and drop-off locations, and choose the rental period and car type. It also allows car rental companies to manage reservation requests, track rental bookings, and schedule car delivery and pickup.

4. ***Payment management:*** This module enables customers to make secure payments. It also allows car rental companies to manage payment processing and generate invoices and receipts.

5. ***Reporting and analytics:*** This module enables car rental companies to generate reports and analytics on car inventory, rental bookings, revenue, and customer feedback. This information can help car rental companies optimize their operations and improve their services.

6. ***Admin management:*** This module enables system administrators to manage user permissions, monitor system activity, and perform system maintenance tasks.

Explanation of a Module: Let me explain the functionality of one of the modules for an Online Car Rental System,

The Car Management Module

The Car Management module is a critical component of the system that enables car rental companies to manage their car inventory efficiently. The module includes several features, such as:

1. ***Car information management:*** This feature allows car rental companies to add new cars to their inventory, update existing car information, and remove cars that are no longer available for rent. Car rental companies can also add images of the cars, including interior and exterior photos, to provide customers with a more detailed view of the cars available for rent.
2. ***Car availability management:*** This feature enables car rental companies to manage the availability of their cars, including scheduling maintenance and repairs and blocking out specific rental periods.
3. ***Car classification:*** This feature allows car rental companies to categorize their cars into different classes, such as economy, compact, midsize, luxury, and SUV. This feature makes it easier for customers to choose the car that best fits their needs and preferences.
4. ***Pricing management:*** This feature enables car rental companies to set pricing for their cars, including daily, weekly, and monthly rental rates. Car rental companies can also set different pricing for different car classes and offer discounts and promotions to attract customers.
5. ***Reporting and analytics:*** This feature enables car rental companies to generate reports and analytics on their car inventory, including the number of cars available for rent, the number of rentals per car, and revenue generated per car. This information can help car rental companies optimize their car inventory and pricing strategies to maximize profitability.

System Limitations/Constraints

The Online Car Rental System, like any other system, has some limitations and constraints. Here are some of the common limitations and constraints that car rental companies may face when implementing an Online Car Rental System:

A. ***Internet connectivity:*** The system is entirely dependent on internet connectivity. If the internet connection is lost, the system may not function correctly, and customers may not be able to make bookings or payments.

B. ***Technical issues:*** Technical issues such as system downtime or software bugs can occur, causing inconvenience to customers and car rental companies.

C. ***Human error:*** Mistakes made by staff members, such as entering incorrect data, can cause issues for customers and car rental companies.

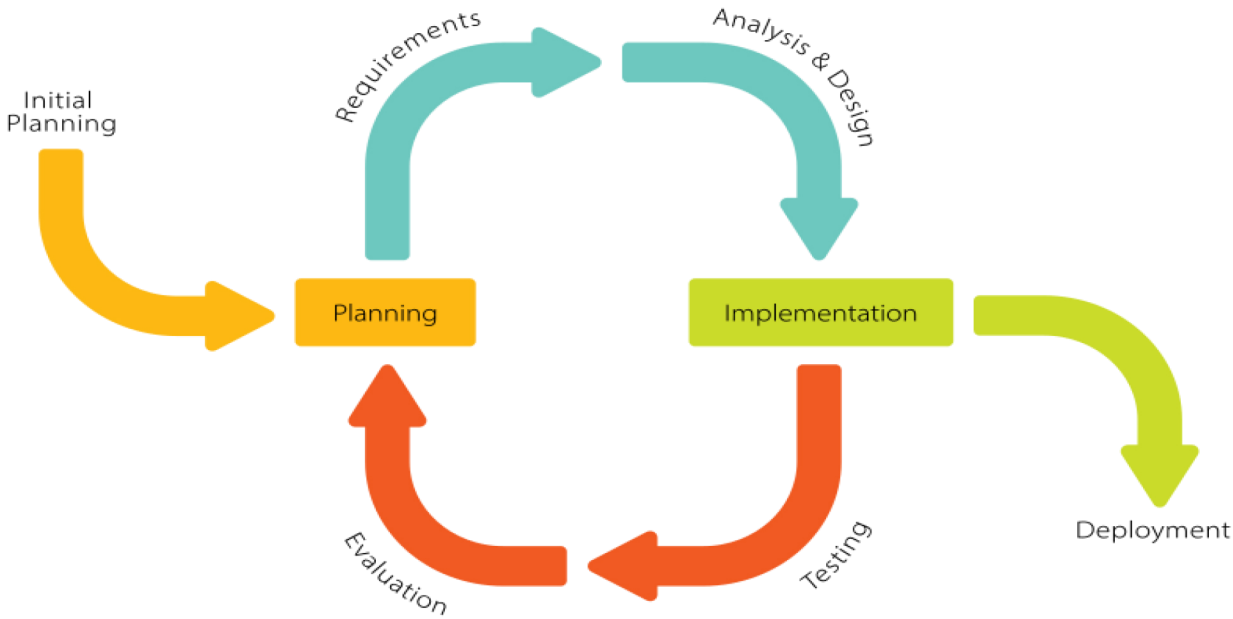
D. ***System cost:*** Implementing and maintaining an Online Car Rental System can be costly, especially for small car rental companies that may not have the resources to invest in expensive technology.

Despite these limitations and constraints, the Online Car Rental System can provide significant benefits to car rental companies and customers alike. By addressing these limitations and constraints proactively, car rental companies can minimize their impact and ensure the smooth functioning of the system.

Software Process Methodology

Iterative Model: The iterative model is an approach to software development that involves breaking down the development process into smaller, incremental iterations. Each iteration consists of a set of activities, such as requirements gathering, design, implementation, testing, and evaluation. The iterative model allows for the repetition of these activities in multiple cycles, with each cycle building upon the previous one.

Diagrammatically



When applying the iterative model to the development of a car rental system, the following steps can be followed:

Requirements Gathering: In this initial phase, the requirements for the car rental system are identified by interacting with stakeholders, including car rental company representatives, customers, and system administrators. The requirements can include features such as user registration, vehicle inventory management, reservation handling, billing, and reporting.

Designing: Based on the gathered requirements, the system's architecture and design are created. This includes designing the user interface, database structure, and system components. The design should accommodate future iterations and enhancements.

Implementation: The implementation phase involves coding the software based on the design specifications. The focus is on developing core functionalities and modules that fulfill the immediate requirements of the car rental system.

Iterations: Based on the feedback and evaluation results, the development phase move to the next iteration. The subsequent iterations follow the same cycle of requirements gathering, design, implementation, testing, evaluation, and feedback.

Testing: After the initial implementation, testing is conducted to identify and fix any defects or issues. Different types of testing, such as unit testing, integration testing, and system testing, are performed to ensure the system works as intended.

Evaluation and Feedback: Once the system has been tested, it is evaluated by stakeholders, including end-users and car rental company representatives. Feedback is gathered to assess the

system's functionality, usability, and performance. This feedback helps identify areas for improvement and guides the next iteration.

Tools and Technologies

The Online Car Rental System can be developed using a variety of tools and technologies, including: Programming Languages, Web Frameworks, Front-end Frameworks etc. can be used to develop the user interface of the Online Car Rental System.

Table 2 Tools and Technologies for Proposed Project

	Tools	**Rationale**
Tools And Technologies	MS Visual Studio	Coding / Editor
	My SQL	DBMS
	MS Word	Documentation
	MS Power Point	Presentation
	Technology	**Rationale**
	Xamp Server	Web hosting / server
	SQL	Query Language
	Html, CSS, Bootstrap	Web Development
	PHP	Server Language

Concept

A rental car system is a service that allows customers to rent cars for temporary use. The concept typically involves a company or organization that owns a fleet of vehicles that customers can rent for a specified period of time, usually ranging from a few hours to several days or weeks.

The rental car system typically operates through a website that allows customers to browse available vehicles, select the dates and times they need the car, and make a reservation. The system also include features such as the ability to choose pick-up and drop-off locations, select add-ons like car seats.

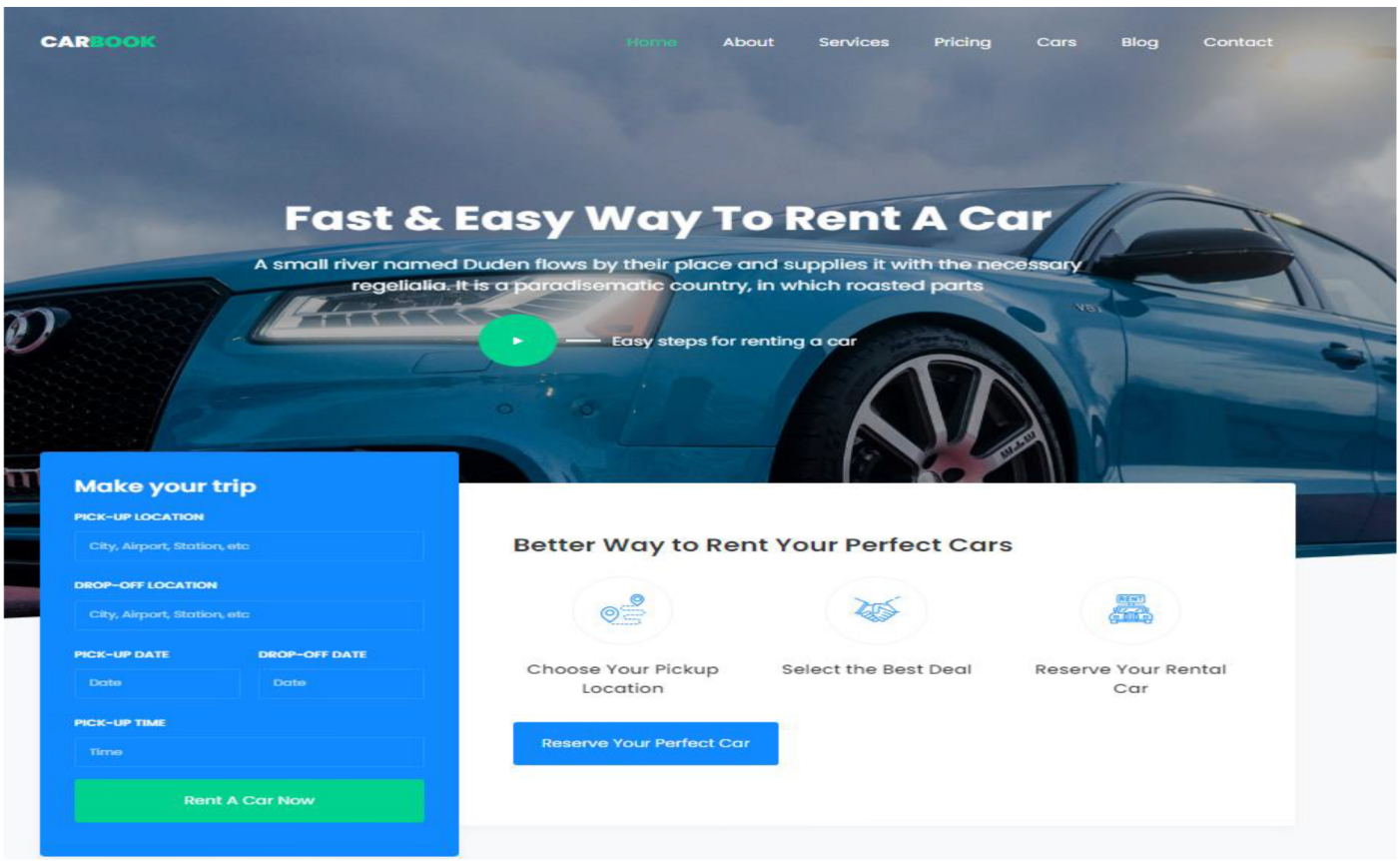
Once the reservation is confirmed, customers can pick up the rental car at the designated location and return it at the end of the rental period. The rental car company may inspect the vehicle for any damages or issues before and after the rental, and customers may be responsible for any damage or cleaning fees incurred during their (self-driving) use of the car.

Overall, a rental car system provides a convenient and flexible transportation option for individuals who need a temporary vehicle for travel, work, or other purposes.

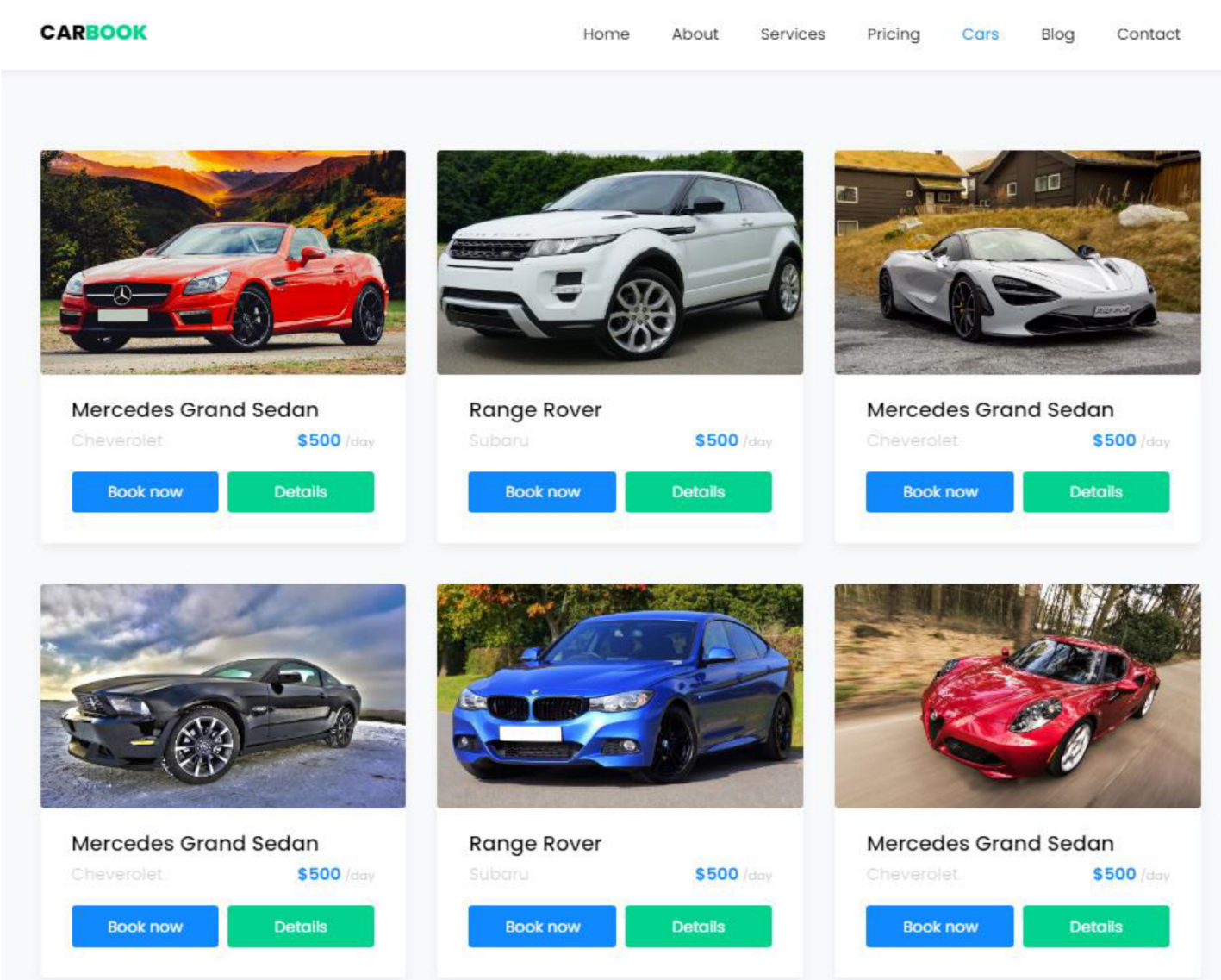
Mockups

Some guidance on what a mockup for a car rental service might include:

Home Page: The home page should include a search bar where we can enter their pick-up and drop-off locations, dates, and times. It should also include information about the rental company, available vehicles, and any promotions or deals.



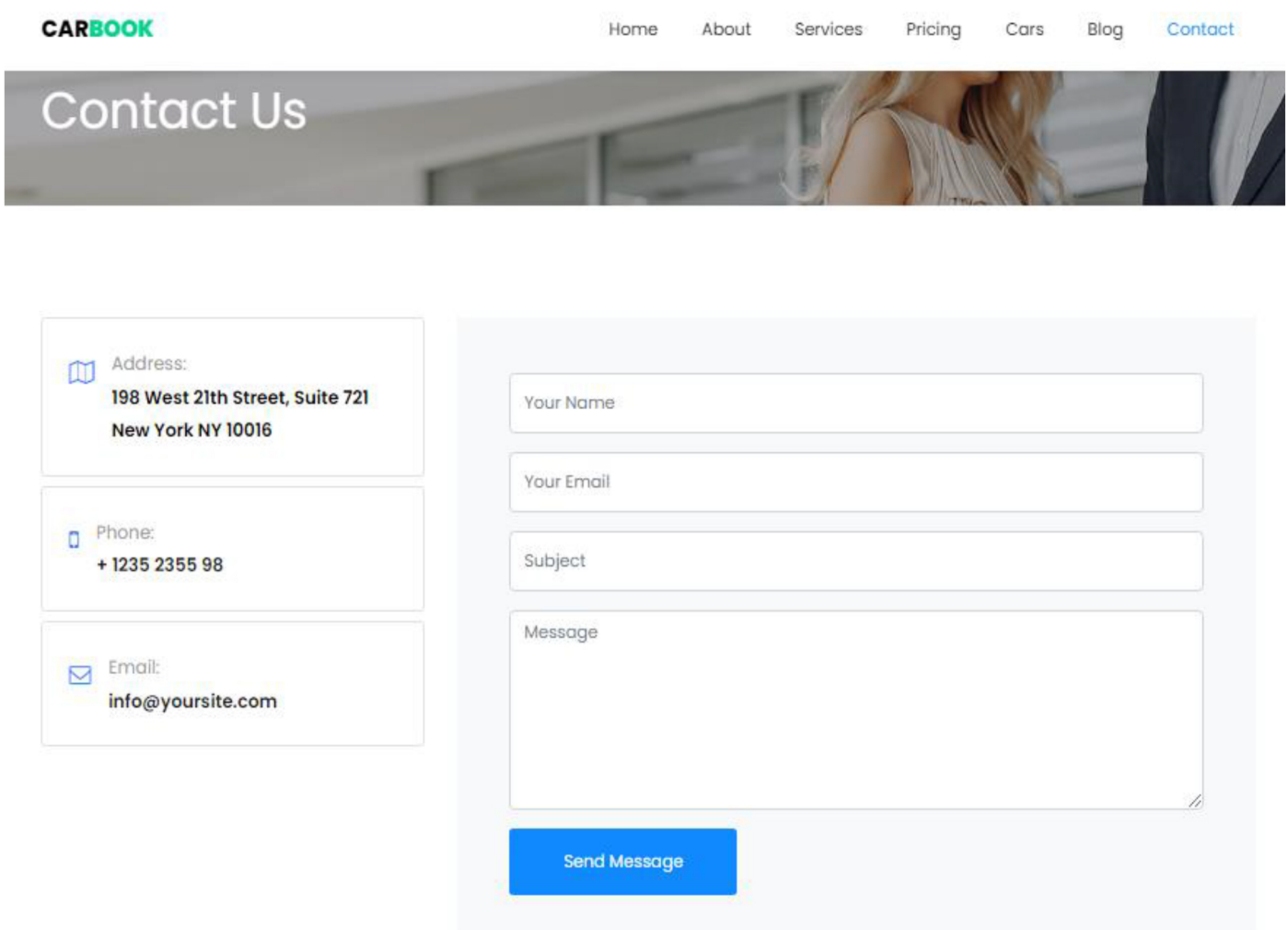
Vehicle Selection Page: This page should display all available vehicles based on the search criteria entered on the home page. It should include photos of each vehicle, along with details such as make, model, year, features, and price per day. Customers should be able to filter the results based on their preferences.



Booking Page: Once the customer has selected a vehicle, they should be taken to a booking page where they can enter their personal information, driver's license details, and credit card information. They should also be able to select any add-ons they want, such as car seats.

	Per Hour Rate	Per Day Rate	Leasing
 Cheverolet SUV Car rated: ★★★★☆	$10.99 /per hour $3/hour fuel surcharges	$60.99 /per day $3/hour fuel surcharges	$995.99 /per month $3/hour fuel surcharges
 Cheverolet SUV Car rated: ★★★★☆	$10.99 /per hour $3/hour fuel surcharges	$60.99 /per day $3/hour fuel surcharges	$995.99 /per month $3/hour fuel surcharges
 Cheverolet SUV Car rated: ★★★★☆	$10.99 /per hour $3/hour fuel surcharges	$60.99 /per day $3/hour fuel surcharges	$995.99 /per month $3/hour fuel surcharges

Customer Support Page: This page should include information on how to contact customer support, as well as answers to frequently asked questions about the rental process.



These are just a few examples of the pages that a car rental service might include in their mockups. (**The specific design and layout will depend on the company's branding and target audience)**

Conclusion

Car rental business has emerged with a new goodies compared to the past experience where every activity concerning car rental business is limited to a physical location only [3]. Even though the physical location has not been totally eradicated; the nature of functions and how these functions are achieved has been reshaped by the power of internet [6]. Nowadays, customers can reserve cars online, rent car online, and have the car brought to their door step once the customer is a registered member or go to the office to pick the car [3]. The web based car rental system has offered an advantage to both customers as well as Car Rental Company to efficiently and effectively manage the business and satisfies customers need at the click of a button [3].

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